

DENTAL CALCULUS

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FINAL YEAR PART 1

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DEFINITION

According to Carranza, Calculus consists of mineralized bacterial plaque that forms on the surfaces of natural teeth and dental prostheses.

Dental calculus is an adherent, calcified or calcifying mass that forms on the surfaces of teeth and dental appliances. It is covered on its external surface by vital, tightly adherent, non-mineralized plaque.

CLASSIFICATION

Dental calculus is classified on the basis of its location.

1. Supragingival calculus
2. Subgingival calculus

SUPRAGINGIVAL CALCULUS

- It is located coronal to the gingival margin and therefore is visible in the oral cavity.
- It is usually white or whitish yellow in color, hard, with a clay-like consistency and easily detached from the tooth surface.
- The two most common locations: buccal surfaces of maxillary molars and lingual surfaces of mandibular anterior teeth



Fig 1: Supragingival calculus on the buccal surfaces of maxillary molars adjacent to the orifice for the parotid duct



Fig 2: Extensive supragingival calculus is present on the lingual surfaces of the lower anterior teeth

SUBGINGIVAL CALCULUS

- It is located below the crest of the marginal gingiva and therefore is not visible on routine clinical examination.
- It is typically hard and dense and appears to be dark brown or greenish black in color and it is firmly attached to the tooth surface

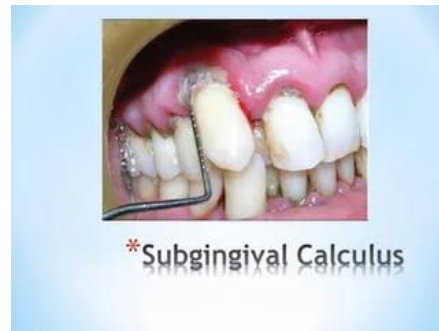


Fig 3: Subgingival calculus

DIFFERENCES BETWEEN SUPRAGINGIVAL AND SUBGINGIVAL CALCULUS

SUPRAGINGIVAL CALCULUS

- Located coronal to the gingival margin-visible in the oral cavity.
- White or whitish yellow in color.
- Hard with clay-like consistency.
- Easily detached from the tooth surface.
- Source-derived from salivary secretions

SUBGINGIVAL CALCULUS

- Located below the crest of the marginal gingiva-not visible on routine clinical examination
- Dark brown or greenish black in color.
- Hard and dense.
- Firmly attached to the tooth surface.
- Formed from gingival exudate.

COMPOSITION

SUPRAGINGIVAL CALCULUS

Inorganic content

- Calcium phosphate – 75.9%
- Calcium carbonate – 3.1%
- Traces of magnesium phosphate

Two-thirds of inorganic content is crystalline in nature:

- Hydroxyapatite – 58%
- Consists of a mixture of protein - Polysaccharide complexes, desquamated epithelial cells, leukocytes, & various

Organic content

- Magnesium whitlockite -21%
- Octacalcium phosphate-12%
- Brushite -9%

Types of microorganisms.

- Carbohydrates: 1.9-9.1%
- Salivary proteins: 5.9-8.2%
- Lipids: 0.2%

SUBGINGIVAL CALCULUS

- Ratio of calcium to phosphate is higher subgingivally.
- Similar hydroxyapatite content
- More magnesium whitlockite compared to supragingival calculus
- Less brushite and octacalcium phosphate than supragingival calculus.
- Salivary proteins not found subgingivally.

ATTACHMENT TO THE TOOTH SURFACE

Four modes of attachment :

- Attachment by means of an organic pellicle on cementum.
- Mechanical locking into surface irregularities , such as caries lesions or resorption lacunae.
- Close adaptation of the undersurface of calculus to depressions or gently sloping mounds

of the unaltered cementum surface.

- Penetration of bacterial calculus into cementum.

CALCULUS FORMATION

- Calculus is mineralized dental plaque.
- Soft plaque is hardened by precipitation of mineral salts, which usually starts between the 1st and 14th days of plaque formation. Calcification has been reported to occur within as little as 4-8 hours.
- Calcifying plaques may become 50% mineralized in 2 days and 60-90% mineralized in 12 days.
- Early plaque contains a small amount of inorganic material, which increases as the plaque develops into calculus.
- Plaque that does not develop into calculus reaches a plateau of maximal mineral content within 2 days.
- Calcification requires binding of calcium ions to the carbohydrate-protein complexes of the organic matrix and the precipitation of crystalline calcium phosphate salts.
- Calculus formation continues until it reaches a maximum from which it may be reduced in amount.
- The time required to reach the maximal level has been reported to be between 10 weeks and 6 months.
- The decline from maximal calculus accumulation – reversal phenomenon.

CHANGES WHILE CALCIFICATION

- As calcification progresses, the number of filamentous bacteria increases, and the foci of calcification changes from basophilic to eosinophilic.
- Calculus is formed in layers, which are often separated by a thin cuticle that becomes embedded in the calculus as calcification progresses.

FACTORS AFFECTING CALCULUS DEPOSITION

- Diet and nutrition: Increased calculus formation has been associated with deficiencies of vitamin A, niacin, or pyridoxine, and with an increase in dietary calcium, phosphorus, bicarbonate, protein and carbohydrates.
- Age: There is an increase in calculus deposition with increasing age due to change in quantity and quality of saliva with age, favoring the mineralization properties.
- Habits: In populations that practice regular oral hygiene and with access to regular professional care have a low tendency for calculus formation.
- Smoking: is associated with an elevated risk for supragingival calculus deposition. Smoking may exert its influence systemically (elevated levels of salivary calcium and phosphorus) or locally via a conditioning of tooth surfaces.
- Salivary pH: increase pH, increases the calculus formation.
- Salivary flow rate: increased salivary flow rate decreases the calculus formation. Salivary flow rate affects calcium phosphate saturation.
- Salivary calcium concentration: Elevated salivary calcium concentration increases the rate of calculus formation.
- Emotional status: increased calculus formation has been associated with disturbed emotional status.
- Nucleation inhibitors: Mg blocks apatite crystallization and stabilizes calcium phosphate as amorphous mineral

THEORIES

- BOOSTER CONCEPT
- EPITACTIC CONCEPT
- INHIBITION THEORY
- TRANSFORMATION THEORY
- BACTERIOLOGICAL THEORY
- ENZYMATIC THEORY

BOOSTER CONCEPT

According to this concept, mineral precipitation results from a local rise in the degree of saturation

of calcium and phosphate ions, which may be brought about in several ways:

- A rise in the pH of the saliva causes the precipitation of calcium phosphate salts by lowering the precipitation constant.
- Colloidal proteins in saliva bind calcium and phosphate ions and maintain a supersaturated solution with respect to calcium phosphate salts.
- Phosphatase liberated from dental plaque, desquamated epithelial cells, or bacteria precipitates calcium phosphate by hydrolyzing organic phosphates in saliva, thereby increasing the concentration of free phosphate ions.

EPITACTIC CONCEPT/ SEEDING THEORY/ HETEROGENEOUS NUCLEATION

- Seeding agents induce small foci of calcification that enlarge and coalesce to form a calcified mass. This concept referred to as epitactic concept or heterogeneous nucleation.
- The seeding agents in calculus formation are not known, but it is suspected that the intercellular matrix of plaque plays an active role.
- The carbohydrate-protein complexes may initiate calcification by removing calcium from the saliva (chelation) and binding with it to form nuclei that induce the subsequent deposition of minerals.

INHIBITION THEORY

- This theory consider the possibility of calcification occurring only at special site because there exist an inhibiting mechanism at non-calcifying sites.
- Wherever calcification occur the inhibitor is either altered or removed.
- One such inhibiting agent could be polysulphate, which prevent the initial nucleus from growing by possibly 'poisoning' the growth center of crystal.

TRANSFORMATION THEORY

- Hypothesis – hydroxyapatite need not arise exclusively via epitaxis or nucleation.
- Amorphous non-crystalline deposits and brushite —> Transformed into octacalcium phosphate and then to hydroxyapatite.
- Controlling mechanism in transformation mechanism can be pyrophosphate.

BACTERIOLOGICAL THEORY

- Oral microorganisms are the primary cause of calculus formation.
- Involved in the attachment to the tooth surface.
- Leptotrichia and Actinomyces have been considered most often as the causative microorganism.

ENZYMATIC THEORY

Calculus formation → Action of phosphatases derived from either oral tissues or oral microorganism on some salivary phosphate complex.

ROLE OF MICROORGANISMS IN CALCULUS MINERALIZATION

- The role of bacteria in calculus formation is apparently complex. Although the formation of calculus in germ – free animals indicates that bacteria are not essential, but when present they obviously play an important role.
- Active → Forming phosphatases , Increased plaque PH , Induce mineralization.
- Passive → Bacteria are simply calcified along with other plaque components.
- Mineralization occurs both extracellularly and intracellularly around gram-positive and gram-negative organisms.
- Filamentous organisms , diphtheroids , and Bacterionema and Veillonella spp., have the ability to form intercellular apatite crystals.
- Calculus formation spreads until the matrix and bacteria are calcified.

ETIOLOGIC SIGNIFICANCE

- The incidence of calculus, gingival inflammation and periodontal disease increases with age.
- Calculus does not contribute directly to gingival inflammation, but it provides a fixed nidus for the continued accumulation of plaque and its retention in close proximity to the gingiva.
- Plaque initiates gingival inflammation, which leads to pocket formation.
- The bacterial plaque that coats the teeth is the main etiologic factor in the development of periodontal disease, the removal of subgingival plaque and calculus constitutes the

cornerstone of periodontal therapy.

ANTICALCULUS AGENTS

First generation agents:

- Dissolution agents: Agents such as acids, spring salts, and sodium ricinoleate.
- Plaque attachment inhibitors: Inhibitors like silicones and ion exchange resins.
- Plaque inhibition agents: Agents like antibiotics, antiseptics, eg: chloramine-T
- Matrix disruption agents: Agents such as enzymes, ascorbic acid, sodium percarbonate, copper sulfite, and 30% urea.

Second generation agents:

- Inhibitors of crystal growth: Vitamin C, pyrophosphate, diphosphonate, zinc salts, calcium lactate, sodium fluoride, copolymer, citroxain and sodium citrate.

MATERIAL ALBA

- The accumulation of microorganisms desquamated epithelial cells, leukocytes and a mixture of salivary proteins and lipids with few or no food particles and lacks a regular internal pattern observed in plaque.
- Yellow or greyish white, soft sticky deposit and is somewhat less adherent than dental plaque.



Fig 4: Material alba

FOOD DEBRIS

- Most food debris is liquefied by bacterial enzymes and cleared from the oral cavity by salivary flow and mechanical action of tongue, cheek, and lips.

- Aqueous solutions are typically cleared within 15 minutes ,wheras sticky foods may adhere more than 1 hour.



Fig 5: Food debris

DENTAL STAINS

- Pigmented deposits on the tooth surface are called dental stains.
- Stains are primarily and aesthetic problem and do not cause inflammation of the gingiva.
- The use of tobacco products , coffee, tea , certain mouth rinses and pigments in food and contribute to stain formation.



Fig 6: Dental stains

CONCLUSION

- Calculus plays an important role in maintaining and accentuating periodontal disease by keeping plaque in close contact with the gingival tissues and creating areas where plaque removal is impossible.
- Therefore the clinician must not only possess the clinical skills to remove the calculus and other irritants that attach to teeth.

REFERENCE

- NEWMAN AND CARRANZA'S CLINICAL PERIODONTOLOGY – THIRD SOUTH ASIA EDITION
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